

PROJECT REPORT

EXECUTIVE SUMMARY

The Mutual Fund Analytics Platform was developed as part of the Bluestock Fintech Capstone Project to address the growing need for data-driven investment analysis in India's rapidly expanding mutual fund industry. With industry Assets Under Management (AUM) exceeding ₹81 lakh crore, over 26 crore investor folios, and record SIP inflows surpassing ₹31,000 crore, investors and financial advisors require a centralized platform capable of transforming complex financial data into actionable insights.

The primary objective of this project was to design and implement an end-to-end data analytics solution that integrates multiple mutual fund datasets, automates data processing through ETL pipelines, performs advanced financial analysis, and presents key insights through an interactive business intelligence dashboard. The platform consolidates data related to mutual fund schemes, NAV history, AUM trends, SIP inflows, investor transactions, benchmark indices, and portfolio holdings into a unified analytical framework.

A robust ETL pipeline was developed using Python and Pandas to extract, clean, validate, transform, and load data from multiple sources. The processed datasets were structured to support efficient analytics and reporting. Exploratory Data Analysis (EDA) was conducted to identify industry trends, investor behaviour patterns, category-wise fund flows, geographic investment distribution, and fund performance dynamics.

To evaluate fund performance comprehensively, several financial metrics were calculated, including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, Value at Risk (VaR), Conditional Value at Risk (CVaR), and Rolling Sharpe Ratios. These metrics enabled risk-adjusted comparisons across mutual fund schemes and benchmark indices. A fund scorecard framework was also developed to rank schemes based on return potential, risk characteristics, and consistency of performance.

Advanced analytics techniques were applied to generate deeper insights. Investor cohort analysis was used to study investment behaviour across different investor groups, SIP continuity analysis identified potentially at-risk investors, and sector concentration analysis measured portfolio diversification using the Herfindahl-Hirschman Index (HHI). Additionally, a rule-based recommendation engine was developed to suggest suitable mutual funds based on investor risk appetite categories such as Low, Moderate, and High.

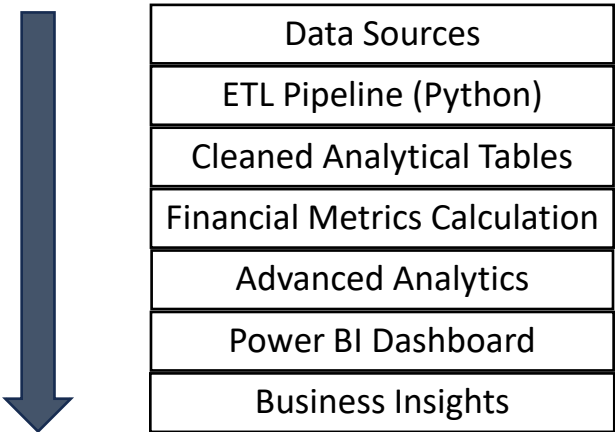
To support business decision-making, an interactive Power BI dashboard was created consisting of four analytical modules: Industry Overview, Fund Performance, Investor Analytics, and SIP &

Market Trends. The dashboard includes KPI cards, drill-through functionality, tooltips, slicers, trend visualizations, and comparative performance analysis, enabling stakeholders to explore mutual fund data dynamically and efficiently.

The project successfully demonstrates the application of data engineering, financial analytics, business intelligence, and risk management concepts in a real-world fintech environment. The resulting platform provides investors, analysts, and financial institutions with a scalable framework for evaluating mutual fund performance, understanding investor behaviour, monitoring industry trends, and supporting informed investment decisions.

Overall, the Mutual Fund Analytics Platform transforms fragmented financial datasets into a unified, insight-driven ecosystem, showcasing how modern data analytics can enhance transparency, efficiency, and decision-making within the mutual fund industry.

PROJECT ARCHITECTURE



Components

Data Layer

- Mutual Fund Schemes
- NAV History
- SIP Data
- Investor Transactions
- Portfolio Holdings

Processing Layer

- Data Cleaning

- Missing Value Treatment
- Feature Engineering

Analytics Layer

- CAGR
- Sharpe Ratio
- Sortino Ratio
- VaR
- CVaR
- Cohort Analysis

Visualization Layer

- Power BI Dashboards

DATA SOURCES & DATASET DESCRIPTION

The Mutual Fund Analytics Platform was built using publicly available datasets obtained from AMFI India, mutual fund APIs, benchmark index data, and simulated investor transaction records. The project integrates multiple datasets covering mutual fund performance, investor behaviour, industry trends, and portfolio composition.

Datasets Used

Dataset	Description
Fund Master	Master information for 40 mutual fund schemes including fund house, category, risk grade, and expense ratio
NAV History	Historical daily NAV data from 2022–2026 for all selected mutual fund schemes
AUM by Fund House	Quarterly Assets Under Management (AUM) data for leading asset management companies
Monthly SIP Inflows	Monthly SIP inflows, active SIP accounts, new registrations, and SIP AUM

Dataset	Description
Category Inflows	Category-wise net inflows across Large Cap, Mid Cap, Small Cap, ELSS, Liquid, and Hybrid funds
Industry Folio Count	Growth of investor folios across the mutual fund industry
Scheme Performance	Fund performance metrics including returns and risk statistics
Investor Transactions	Simulated SIP, Lumpsum, and Redemption transactions for investor behaviour analysis
Portfolio Holdings	Equity holdings, sector allocations, and portfolio weights of mutual funds
Benchmark Indices	Historical benchmark data including Nifty 50, Nifty 100, Midcap, and Small Cap indices

Overall, the project utilized more than 87,000 records spanning approximately 4.5 years of historical data, enabling comprehensive performance analysis and investor analytics.

ETL (EXTRACT, TRANSFORM, LOAD) PIPELINE

A complete ETL pipeline was developed to automate data collection, cleaning, transformation, and loading processes.

Extract

Data was collected from multiple sources including:

- AMFI mutual fund datasets
- Historical NAV records
- Benchmark index datasets
- Investor transaction records
- Portfolio holdings data
- SIP and AUM reports

Python libraries such as Pandas and Requests were used to ingest data from CSV files and APIs.

Transform

Several preprocessing operations were performed:

- Date parsing and standardization
- Missing value treatment
- Duplicate record removal
- Data type correction
- Validation of AMFI codes
- Standardization of transaction categories
- Calculation of derived metrics such as returns and growth rates

The transformation stage ensured data consistency and analytical readiness.

Load

The cleaned datasets were loaded into a structured analytical environment and exported as processed CSV files. The transformed data was further connected to Power BI for dashboard development and visualization.

ETL Outcome

The ETL process successfully converted fragmented mutual fund datasets into a centralized analytical repository, enabling efficient reporting and advanced analytics.

Examples:

Feature	Formula
Annual Return	$(\text{End NAV} / \text{Start NAV}) - 1$
CAGR	Annualized return
Volatility	Std Dev of returns
Sharpe Ratio	Return / Risk

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis was conducted to understand patterns, trends, and relationships within the mutual fund ecosystem.

Analysis Performed

NAV Trend Analysis

Historical NAV movements were analyzed to identify long-term growth trends and market corrections.

AUM Growth Analysis

Quarterly AUM growth across major fund houses was examined to evaluate market leadership and expansion.

SIP Inflow Analysis

Monthly SIP inflows were analyzed to identify growth patterns in retail investor participation.

Category-wise Inflow Analysis

Fund categories were compared to understand investor preferences and capital allocation trends.

Investor Demographic Analysis

Investor behaviour was studied using age groups, geographic locations, city tiers, and income segments.

Geographic Distribution Analysis

State-wise investment participation and transaction volumes were analyzed.

Folio Growth Analysis

The increase in mutual fund folios over time was examined as an indicator of market penetration.

Correlation Analysis

Correlations among fund returns were computed to understand diversification opportunities.

Sector Allocation Analysis

Portfolio holdings were aggregated to identify dominant sectors across mutual fund portfolios.

Key EDA Insights

1. SIP inflows demonstrated a strong upward trend, indicating growing retail participation.
2. Large fund houses continued to dominate industry AUM.
3. Equity-oriented categories attracted the highest investor interest.
4. Younger and middle-aged investors contributed significantly to SIP investments.
5. Investor participation expanded beyond metropolitan cities into emerging markets.
6. Sector allocations showed concentration in Financial Services, Technology, and Banking sectors.
7. Fund returns exhibited varying correlation levels, highlighting diversification benefits.
8. Industry folio counts showed sustained growth throughout the study period.

PERFORMANCE ANALYSIS

Performance analysis was conducted to evaluate mutual fund returns and risk-adjusted performance.

Metrics Calculated

CAGR (Compound Annual Growth Rate)

Used to measure annualized growth over investment periods.

Sharpe Ratio

Evaluated return generated per unit of total risk.

Sortino Ratio

Measured return generated relative to downside risk.

Alpha

Assessed fund outperformance against benchmark indices.

Beta

Measured sensitivity of fund returns relative to market movements.

Maximum Drawdown

Identified the largest historical decline from peak portfolio value.

Value at Risk (VaR)

Estimated potential portfolio loss under normal market conditions.

Conditional Value at Risk (CVaR)

Measured expected loss beyond the VaR threshold.

Fund Ranking Model

A composite scoring framework was developed using:

- 30% Weight: 3-Year Returns
- 25% Weight: Sharpe Ratio
- 20% Weight: Alpha
- 15% Weight: Expense Ratio
- 10% Weight: Maximum Drawdown

This scoring model enabled objective ranking of mutual funds based on both return and risk factors.

Benchmark Comparison

Fund performance was compared against major benchmark indices including:

- Nifty 50
- Nifty 100
- BSE Small Cap

Tracking error and relative performance measures were used to assess consistency and benchmark alignment.

DASHBOARD VISUALIZATIONS

To enable effective decision-making and interactive analysis, a comprehensive Power BI dashboard was developed using the processed mutual fund datasets. The dashboard transforms complex financial and investor data into intuitive visualizations, allowing users to explore industry trends, fund performance, investor behaviour, and SIP market dynamics through an interactive interface.

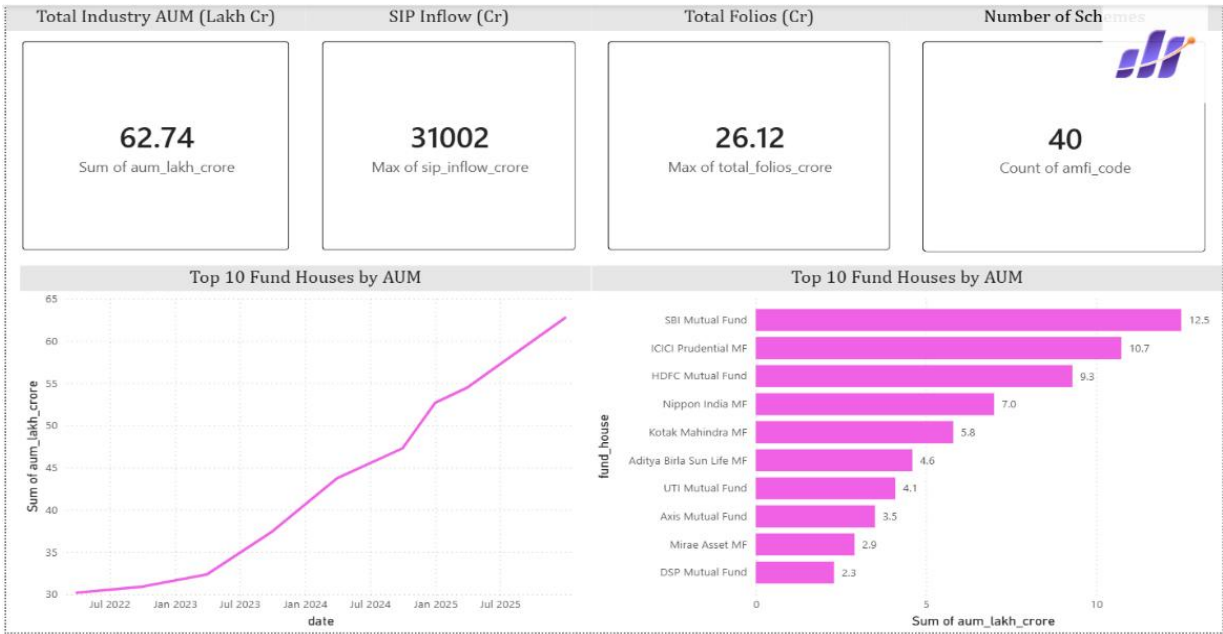
The dashboard consists of four dedicated pages:

- **Industry Overview** – Provides insights into AUM growth, SIP inflows, category-wise fund flows, and overall mutual fund industry trends.
- **Fund Performance Analysis** – Evaluates mutual funds using risk-return analysis, NAV trends, Sharpe Ratio, Alpha, Maximum Drawdown, and overall fund rankings.
- **Investor Analytics** – Examines investor demographics, geographic distribution, transaction patterns, and cohort behaviour.
- **SIP & Market Trends** – Tracks SIP inflows, benchmark index performance, category-wise inflows, and market participation trends.

Interactive slicers, KPI cards, drill-through functionality, conditional formatting, and dynamic visualizations were incorporated to improve user experience and enable detailed exploration of the data.

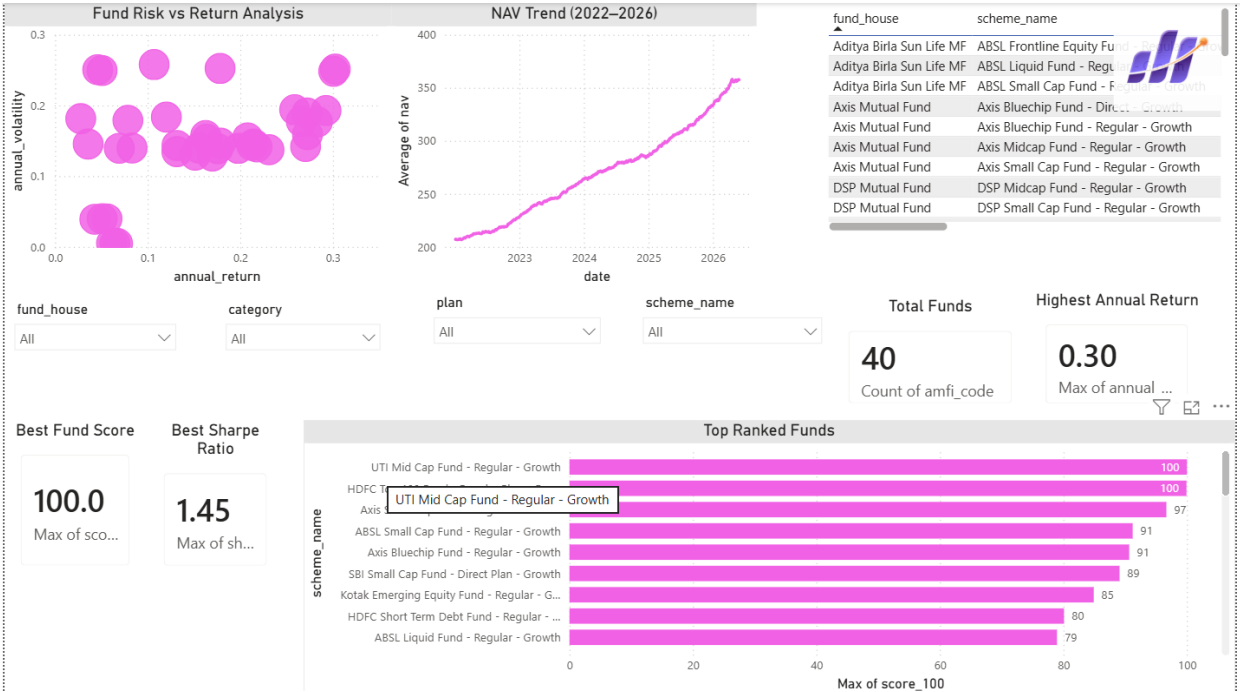
The following screenshots illustrate the key dashboard pages and their analytical capabilities.

Figure 1: Industry Overview Dashboard



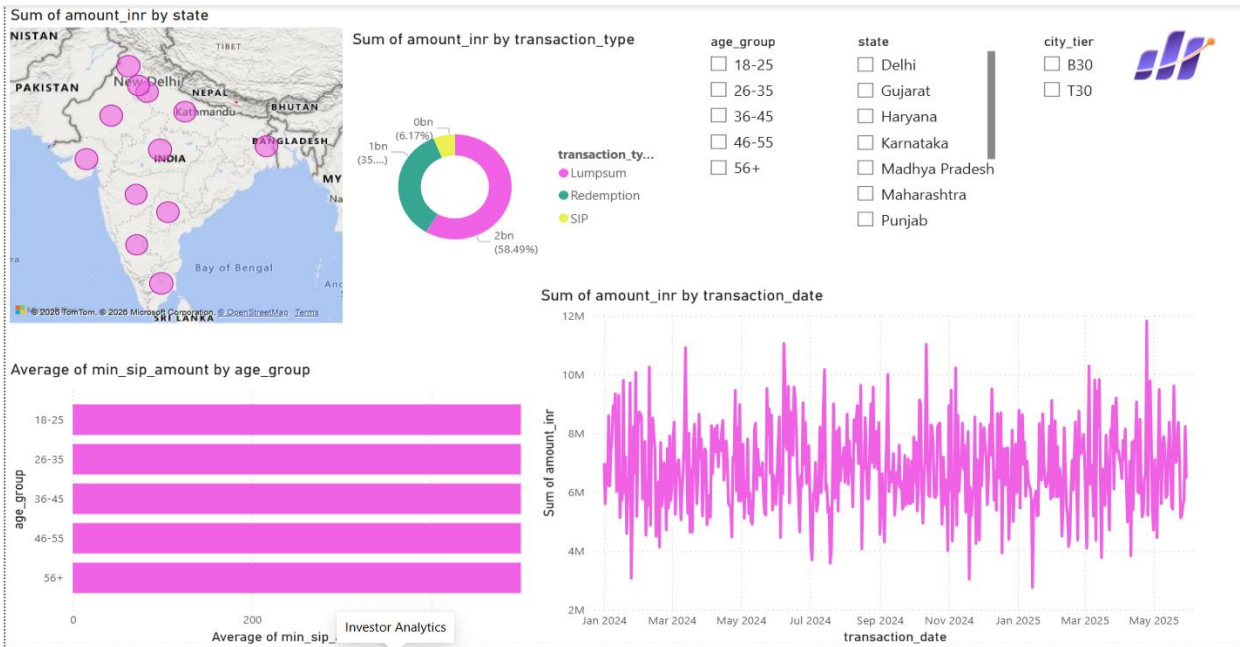
Description: This dashboard provides a high-level overview of the mutual fund industry including AUM trends, SIP inflows, category-wise investments, and industry growth indicators.

Figure 2: Fund Performance Dashboard



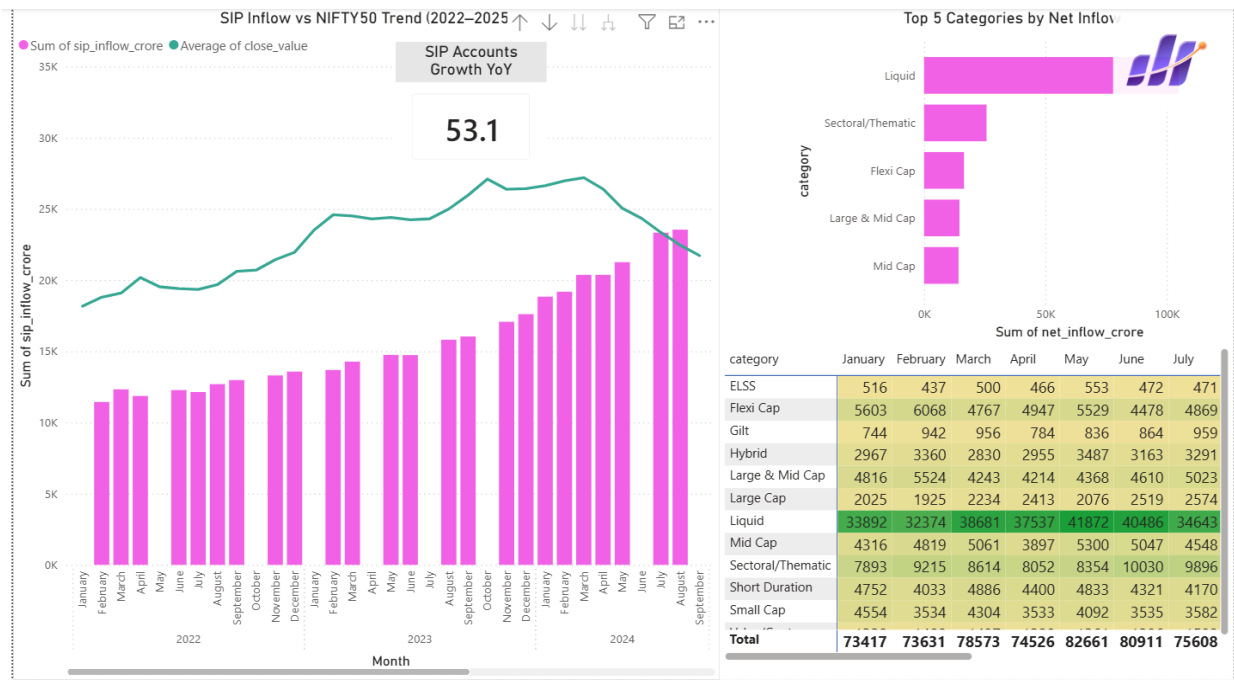
Description: This dashboard evaluates mutual funds based on returns, volatility, risk-adjusted performance metrics, and fund ranking methodology.

Figure 3: Investor Analytics Dashboard



Description: This dashboard analyzes investor demographics, geographic distribution, transaction behaviour, and investment patterns.

Figure 4: SIP & Market Trends Dashboard



Description: This dashboard tracks SIP growth, benchmark index movements, category inflows, and overall market trends

RECOMMENDATIONS

Based on the analysis performed on mutual fund performance, investor behaviour, SIP trends, and risk metrics, the following recommendations are proposed:

1. Promote SIP-Based Investing

The analysis revealed consistent growth in SIP inflows and active SIP accounts over the study period. Asset Management Companies (AMCs) should continue promoting SIP-based investments, particularly among first-time investors, as SIPs encourage disciplined investing and reduce the impact of market volatility through rupee-cost averaging.

2. Improve Investor Diversification

Many investors tend to concentrate investments within a limited number of categories or sectors. Fund managers and advisors should encourage portfolio diversification across Large Cap, Mid Cap, Small Cap, Debt, and Hybrid funds to reduce concentration risk and improve long-term risk-adjusted returns.

3. Utilize Risk-Adjusted Metrics for Fund Selection

Investors should not rely solely on historical returns when selecting mutual funds. Metrics such as Sharpe Ratio, Sortino Ratio, Alpha, Maximum Drawdown, VaR, and CVaR should be considered to evaluate both performance and risk. The developed fund scorecard provides a more balanced approach to fund evaluation.

4. Strengthen Investor Retention Strategies

The SIP continuity analysis identified investors exhibiting irregular investment behaviour and extended transaction gaps. AMCs can implement targeted communication campaigns, reminders, and investor education programs to improve SIP retention and reduce discontinuation rates.

5. Personalize Investment Recommendations

The risk-based recommendation engine demonstrates the potential of personalized investment advisory systems. Financial institutions can further enhance investor experience by recommending schemes aligned with individual risk profiles, investment horizons, and financial goals.

6. Monitor Sector Concentration Risk

Sector concentration analysis highlighted that some portfolios maintain significant exposure to specific sectors. Fund managers should continuously monitor concentration risk and rebalance portfolios when required to maintain diversification and reduce vulnerability to sector-specific downturns.

7. Leverage Business Intelligence Dashboards

Interactive dashboards should be integrated into investment decision-making processes to enable real-time monitoring of fund performance, market trends, investor activity, and risk indicators. Such dashboards improve transparency and support data-driven decision making.

8. Expand Predictive Analytics Capabilities

Future investment platforms can incorporate machine learning models for return forecasting, investor churn prediction, SIP continuation probability, and portfolio optimization to further enhance investment decision support systems.

ADVANCED ANALYTICS METHODOLOGY

Value at Risk (VaR)

Formula:

$\text{VaR}(95\%) = 5\text{th Percentile}(\text{Return Distribution})$

Purpose:

Estimate maximum expected loss under normal market conditions.

Conditional VaR

Formula:

$\text{CVaR} = \text{Mean}(\text{Returns below VaR})$

Purpose:

Estimate extreme downside risk.

Rolling Sharpe Ratio

Formula:

$$\text{Rolling Sharpe} = \frac{\text{Rolling Mean}(\text{Returns})}{\text{Rolling Std}(\text{Returns})} \times \sqrt{252}$$

Purpose:

Evaluate risk-adjusted performance over time.

Cohort Analysis

Investors grouped by first investment year.

Metrics:

- Average SIP
- Total Investment

- Fund Preference

Sector Concentration (HHI)

Formula:

$$HHI = \sum (weight^2)$$

Purpose:

Measure diversification.

Key Findings

Finding 1

SIP inflows increased significantly between 2022 and 2025.

Finding 2

Large fund houses continue to dominate total AUM.

Finding 3

Equity funds remain the preferred category.

Finding 4

Mid-cap and small-cap funds generated superior returns but with higher volatility.

Finding 5

Some funds exhibited high sector concentration risk.

Finding 6

Investor participation is expanding into Tier-2 and Tier-3 cities.

Finding 7

Risk-adjusted ranking differs significantly from return-only ranking.

TECHNOLOGIES USED

Technology Stack

Tool	Purpose
Python	Data Processing
Pandas	ETL & Analytics
NumPy	Financial Calculations
Matplotlib	Visualization
Power BI	Dashboarding
Jupyter Notebook	Development
GitHub	Version Control

LIMITATIONS

While the Mutual Fund Analytics Platform provides comprehensive insights into mutual fund performance and investor behaviour, certain limitations should be acknowledged.

1. Limited Historical Data Horizon

The analysis is based on the available historical data spanning approximately four to five years. Longer historical datasets could provide more robust insights regarding market cycles, economic downturns, and long-term fund performance.

2. Simulated Investor Data

A portion of the investor transaction dataset was generated for educational and analytical purposes. Therefore, investor behaviour patterns observed in the analysis may not fully represent actual market behaviour.

3. Static Risk Assumptions

Risk metrics such as Sharpe Ratio, VaR, CVaR, and Maximum Drawdown were calculated using historical data. These measures assume that future market behaviour will exhibit characteristics similar to historical trends, which may not always hold true.

4. Simplified Recommendation Engine

The recommendation system uses rule-based logic driven primarily by investor risk appetite. In real-world applications, recommendation engines typically consider additional factors such as age, income, financial goals, investment horizon, tax considerations, and market outlook.

5. Data Quality Dependency

The accuracy of analytical outputs depends heavily on the quality and completeness of source datasets. Missing values, reporting inconsistencies, delayed updates, or incorrect records may impact the reliability of the results.

6. Absence of Real-Time Data Integration

The platform operates using batch-processed historical datasets and does not currently support real-time market feeds or live NAV updates. Consequently, dashboard insights reflect historical analysis rather than current market conditions.

7. Limited Benchmark Coverage

The project primarily focuses on major benchmark indices and selected mutual fund schemes. Inclusion of additional indices, international funds, ETFs, and alternative investment products would provide a broader comparative framework.

8. Computational Simplifications

Certain financial calculations and portfolio analytics were simplified to ensure project feasibility within the scope of the capstone program. Professional asset management systems may employ more advanced quantitative models and risk management methodologies.

Future Scope

Future enhancements may include machine learning-based fund recommendation systems, real-time data pipelines, investor churn prediction models, portfolio optimization techniques, sentiment analysis from financial news, and cloud-based dashboard deployment for enterprise-scale applications.

CONCLUSION

The Mutual Fund Analytics Platform successfully demonstrates the application of data engineering, financial analytics, and business intelligence techniques in the fintech domain. By integrating multiple datasets, implementing automated ETL pipelines, calculating advanced risk and performance metrics, and developing interactive Power BI dashboards, the project provides a comprehensive framework for mutual fund analysis.

The platform enables investors and analysts to evaluate fund performance, monitor market trends, analyze investor behavior, and make informed investment decisions using data-driven insights. Advanced analytics such as VaR, CVaR, Rolling Sharpe Ratio, Cohort Analysis, and Sector Concentration Analysis further enhance the depth of financial evaluation.

Overall, the project showcases practical skills in Python, Pandas, Financial Analytics, Data Visualization, and Power BI while delivering a scalable solution that can be extended for real-world investment advisory and portfolio management applications.